

6.8 Solution: From Eq. (6.100), we know that a rotating polarization gives rise to an effective magnetization, with $\mathbf{M} = \mathbf{P} \times \mathbf{v}$, where

$$\mathbf{P} = 3\epsilon_0 E_0 \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \hat{x},$$

and

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = \omega r (\hat{z} \times \mathbf{r}).$$

Then,

$$\begin{aligned} \mathbf{M} &= \mathbf{P} \times \mathbf{v} = 3\epsilon_0 E_0 \omega r \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) [\hat{x} \times (\hat{z} \times \mathbf{r})] \\ &= 3\epsilon_0 E_0 \omega r \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) [\hat{r} \sin \theta \cos \theta \cos \phi - \hat{\theta} \sin^2 \theta \cos \phi], \end{aligned}$$

where we have used $\hat{z} = \hat{r} \cos \theta - \hat{\theta} \sin \theta$ in spherical coordinates. It is straightforward to verify that

$$\nabla \cdot \mathbf{M} = 3\epsilon_0 E_0 \omega \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^3 \sin \theta \cos \theta \cos \phi) - \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin^3 \theta \cos \phi) \right] = 0.$$

Thus, by Eq. (5.100), the magnetic scalar potential can be calculated from surface magnetic charge density,

$$\Phi_m = \frac{3\epsilon_0 E_0 \omega}{4\pi} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \oint \frac{a \sin \theta' \cos \theta' \cos \phi'}{|\mathbf{x} - \mathbf{x}'|} da'.$$

Using the expansion of $|\mathbf{x} - \mathbf{x}'|^{-1}$ in spherical harmonics,

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) Y_{lm}^*(\theta', \phi'),$$

and also notice that

$$\sin \theta' \cos \theta' \cos \phi' = -\frac{1}{2} \sqrt{\frac{8\pi}{15}} (Y_{2,1}(\theta', \phi') - Y_{2,-1}(\theta', \phi')),$$

the surface integral becomes

$$\begin{aligned} & \oint \frac{a \sin \theta' \cos \theta' \cos \phi'}{|\mathbf{x} - \mathbf{x}'|} da' \\ &= 4\pi a^3 \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) \int \left[-\frac{1}{2} \sqrt{\frac{8\pi}{15}} (Y_{2,1}(\theta', \phi') - Y_{2,-1}(\theta', \phi')) Y_{lm}^*(\theta', \phi') \right] d\Omega' \\ &= 4\pi a^3 \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{1}{2l+1} \frac{r_{<}^l}{r_{>}^{l+1}} Y_{lm}(\theta, \phi) \left[-\frac{1}{2} \sqrt{\frac{8\pi}{15}} (\delta_{l,2} \delta_{m,1} - \delta_{l,2} \delta_{m,-1}) \right] \\ &= 4\pi a^3 \cdot \frac{1}{5} \cdot \frac{r_{<}^2}{r_{>}^3} \left[-\frac{1}{2} \sqrt{\frac{8\pi}{15}} (Y_{2,1}(\theta, \phi) - Y_{2,-1}(\theta, \phi)) \right] \\ &= \frac{4\pi a^3}{5} \frac{r_{<}^2}{r_{>}^3} \sin \theta \cos \theta \cos \phi. \end{aligned}$$

Therefore,

$$\begin{aligned}
\Phi_m &= \frac{3\epsilon_0 E_0 \omega}{4\pi} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \frac{4\pi a^3}{5} \frac{r_{<}^2}{r_{>}^3} \sin \theta \cos \theta \cos \phi \\
&= \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 E_0 \omega \frac{a^3}{r_{>}^3} \frac{r_{<}^2}{r^2} \cdot r \sin \theta \cos \phi \cdot r \cos \theta \\
&= \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 E_0 \omega \frac{a^3}{r_{>}^3} \frac{r_{<}^2}{r^2} xz.
\end{aligned}$$

If $r > a$, then $r_{>} = r$, $r_{<} = a$, and

$$\frac{a^3}{r_{>}^3} \frac{r_{<}^2}{r^2} = \left(\frac{a}{r_{>}} \right)^5.$$

On the other hand, if $r < a$, then $r_{>} = a$, $r_{<} = r$, and

$$\frac{a^3}{r_{>}^3} \frac{r_{<}^2}{r^2} = 1 = \left(\frac{a}{r_{>}} \right)^5.$$

Therefore, we can write the final result as

$$\Phi_m = \frac{3}{5} \left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) \epsilon_0 E_0 \omega \left(\frac{a}{r_{>}} \right)^5 xz.$$